

Generation Interconnection System Impact Study Report

For Queue Project Q20-1429

West Farnum 138 kV

February 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1429. The Interconnection Customer proposes a 103.1 MW solar photovoltaic facility interconnecting at the West Farnum 138 kV substation in the Kentucky area. The request was received on March 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 548 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,307,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ubc S Campus - Henry Crk 138kV	Platte City - Albion Road 138kV	151.9	182	21.4
Ubc South Campus - Danz Ave Sw Sta 138kV line	Platte City - Albion Road Substation 138kV L	140.3	192	21.4
Danz Ave Sw Sta - Cwec Tigerton 138kV line	Platte City - Albion Rd Substation 138kV	129.2	202	21.4
Carbon Ctr 500/345kV Transformer	Blackrock - Shawville Sta 345kV Line	124.0	1,998	5.1
Cwec Tigerton - Ellendale 138kV	Albion Road - Platte City 138kV	121.3	209	21.4
Carbon Ctr - Shawville Station 500kV	Shawville Sta - Blackrock 345kV	116.8	2,111	5.1
Shawville Sta - Orchard Road 500kV L	Scott - Shawville Station 345kV Line	112.6	2,944	6.5
Shawville Sta - Blackrock 345kV	Carbon Ctr 500-345 kV Transformer Bank	102.2	2,489	5.9
Silver Bay Power - Hotel Rd 138kV	Coopr - Lee County Solid Waste Energy 345kV	101.2	179	5.5
Coronet - Fort Humbug 500kV	Acadia - Montour Sta 500kV	100.9	3,346	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ubc South Campus - Henry Creek 138kV L	Rebuild line with higher-capacity conductor	\$9,078,000	17% (\$1,583,000)
Ubc South Campus - Danz Ave Sw Sta 138kV line	Rebuild line with higher-capacity conductor	\$15,149,000	17% (\$2,641,000)
Danz Ave Sw Sta - Cwec Tigerton 138kV line	Rebuild line with higher-capacity conductor	\$6,850,000	17% (\$1,194,000)
Carbon Center 500/345kV Transformer Bank	Replace transformer with higher-rated unit	\$36,197,000	3% (\$927,000)
Cwec Tigerton - Ellendale 138kV	Rebuild line with higher-capacity conductor	\$3,528,000	17% (\$615,000)
Carbon Ctr - Shawville Station 500kV	Rebuild line with higher-capacity conductor	\$159,517,000	3% (\$4,087,000)
Shawville Sta - Orchard Road 500kV L	Rebuild line with higher-capacity conductor	\$132,700,000	2% (\$2,963,000)
Shawville Sta - Blackrock 345kV L	Reconductor line	\$18,091,000	2% (\$403,000)
Silver Bay Power - Hotel Rd 138kV	Reconductor line	\$7,323,000	4% (\$265,000)
Coronet - Fort Humbug 500kV	Reconductor line	\$78,954,000	2% (\$1,629,000)
	Total allocated to project		\$16,307,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.